

# Taminul Islam

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## Summary

Machine Learning PhD researcher specializing in efficient LLMs/VLMs, multimodal multi-task learning, and multi-agent systems with 34 peer-reviewed publications and 500+ citations. Built production-ready transformer models achieving 99.3% accuracy on 203K+ image datasets, with 1 paper accepted to WACV 2026, and 2 papers to ICCV 2025. Proven track record delivering lightweight AI solutions (1.38M-5M parameters) outperforming state-of-the-art baselines while optimizing inference speed and memory efficiency. Eligible for F-1 OPT/CPT internships.

## Technical Skills

**Languages & Frameworks:** Python, TensorFlow, PyTorch, Keras, NumPy, Scikit-learn, LoRA/QLoRA, Quantization, JavaScript

**Computer Vision & Multimodal AI:** Semantic Segmentation, Vision Transformers (ViT, Swin), VLM, SAM-2, YOLO, OpenCV

**ML Specializations:** Efficient LLMs/VLMs, Multi-Task Learning, Multi-Agent Systems, Model Optimization, Parameter-Efficient Fine-Tuning

**Data & Analytics:** SQL (PostgreSQL, MySQL), Power BI, Tableau, Excel, Data Visualization, Statistical Analysis

**Tools & Platforms:** Git/GitHub, Google Cloud Platform, Gemini API, Android Debug Bridge (ADB), Docker, LATEX

## Education

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| <b>Southern Illinois University Carbondale</b> , PhD in Computer Science          | Jan 2024 – Present  |
| • GPA: 4.0/4.0   Graduate Research Assistant in AI/ML/Computer Vision at BASE Lab | Carbondale, IL      |
| <b>Daffodil International University</b> , BS in Computer Science and Engineering | Jan 2018 – Dec 2021 |
| • GPA: 3.52/4.0   Full-Free Scholarship (Top 2% of applicants)                    | Dhaka, Bangladesh   |

## Experience

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| <b>Research Assistant – AI/ML/Computer Vision</b> , BASE Lab, SIUC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jan 2024 – Present  |
| <ul style="list-style-type: none"><li>Developed FUME, a dual-stream cross-modal fusion network for automated rumen acidosis detection from OGI thermal camera data; achieved multi-task segmentation (CO<sub>2</sub>/CH<sub>4</sub>) and 3-class pH classification with cross-modal attention mechanism; accepted to WACV 2026 [<a href="https://github.com/taminulislam/fume">github.com/taminulislam/fume</a>]</li><li>Co-developed GasTwinFormer, a hybrid vision transformer for livestock methane detection, processing optical gas imaging data for dietary classification; achieved 92% segmentation accuracy and accepted to ICCV 2025 (Oral, top 1.5% of submissions) [<a href="https://github.com/toqitahamid/gastwinformer">github.com/toqitahamid/gastwinformer</a>]</li><li>Architected CarboFormer, a lightweight semantic segmentation model (5.07M parameters) achieving 84.88% mIoU for CO<sub>2</sub> emission quantification—15% improvement over baseline with 3x faster inference speed; accepted to ISVC 2025 (Oral presentation) [<a href="https://github.com/taminulislam/carboformer">github.com/taminulislam/carboformer</a>]</li><li>Built WeedSwin hierarchical vision transformer achieving 99.3% accuracy in multi-stage weed detection across 203,567 images; published in Scientific Reports (Nature, Q1 journal) and featured in WeedSense multi-task architecture for ICCV 2025 (Poster) [<a href="https://github.com/taminulislam/weedswin">github.com/taminulislam/weedswin</a>]</li></ul> |                     |
| <b>Executive</b> , ServEngineBPO Ltd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Aug 2022 – Nov 2023 |
| <ul style="list-style-type: none"><li>Analyzed system performance data using SQL and Excel to optimize IT infrastructure for 50+ users; built automated reporting dashboards tracking digital advertising metrics, reducing system downtime by 40% through data-driven decision-making</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |
| <b>Research And Development Assistant</b> , Quadque Technologies Ltd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Feb 2022 – Jun 2022 |
| <ul style="list-style-type: none"><li>Built predictive analytics models using Python, SQL, and Scikit-learn for client applications; performed feature engineering, and statistical analysis; created Tableau dashboards communicating insights to business stakeholders</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |
| <b>Undergraduate Research Assistant – Machine Learning</b> , Daffodil International University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Apr 2020 – Jan 2022 |
| <ul style="list-style-type: none"><li>Published 15+ peer-reviewed papers on cybersecurity, NLP, and medical imaging using TensorFlow, PyTorch, and NumPy; developed breast cancer prediction model with 97% accuracy using XGBoost on 500-patient dataset (Scientific Reports, 55+ citations)</li><li>Engineered deep learning pipelines for disaster tweet classification, cardiovascular disease prediction, and fake news detection, achieving 90%+ accuracy across all domains with optimized hyperparameter tuning</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |

## Projects

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| <b>Mobile QA Multi-Agent System</b>   Python, Gemini API, ADB, Google ADK [ <a href="https://github.com/taminulislam/mobile-qa-agent">github.com/taminulislam/mobile-qa-agent</a> ]                                                                                                                            |
| <ul style="list-style-type: none"><li>Engineered multi-agent testing framework with Supervisor-Planner-Executor architecture for automated mobile QA; integrated vision-language model (Gemini 2.5 Flash) for screenshot interpretation and natural language test case execution on Android emulator</li></ul> |

- Implemented differentiated failure detection system achieving accurate reporting of execution vs. assertion failures; demonstrated modular LLM integration enabling seamless substitution of language models across agent roles

**FUME: Fused Unified Multi-Gas Emission Network** | *PyTorch, Cross-Modal Attention, Multi-Task Learning* |

[ [github.com/taminulislam/fume](https://github.com/taminulislam/fume)]

- Designed dual-stream ResNet-50 architecture with cross-modal attention for CO<sub>2</sub>/CH<sub>4</sub> gas emission analysis from thermal OGI camera data; implemented multi-task learning for simultaneous pixel-level segmentation and pH-based acidosis classification (Healthy/Transitional/Acidotic) on 8,967-sample dataset with modality dropout and Focal Loss for 52% class imbalance; accepted to WACV 2026

**EfficientMTNet: Lightweight Brain Tumor Multi-Task Network** | *PyTorch, Deep Supervision* |

[ [github.com/taminulislam/EfficientMTNet](https://github.com/taminulislam/EfficientMTNet)]

- Developed ultra-compact multi-task learning architecture (1.38M parameters) for simultaneous brain tumor classification and segmentation, achieving 99.07% accuracy, 76.25% mIoU, and 1124.5 FPS inference speed—98.5% smaller than Vision Transformer with highest normalized efficiency score (1.000); deployed deep supervision strategy for enhanced gradient flow in resource-constrained medical imaging

**MESA: Medical LLM Safety & Accuracy Evaluation** | *PyTorch, LoRA/QLoRA, 4-bit Quantization* |

[ [github.com/taminulislam/GenAI\\_MESA](https://github.com/taminulislam/GenAI_MESA)]

- Evaluated 3 fine-tuned medical LLMs (117M-2.7B parameters) across 20,000 samples using 16 metrics spanning accuracy, safety, and efficiency; applied parameter-efficient fine-tuning (LoRA) training only 0.1-1% of parameters with 4-bit quantization for 75% memory reduction
- Optimized BioGPT-Large achieving 67.2% exact match accuracy and 78.1% medical entity recognition for clinical deployment; demonstrated DialoGPT-Small 100% safety rate with 0.22 GB GPU memory for patient-facing applications on RTX 3090

## Selected Publications (34 total, 500+ citations, h-index: 14)

- **Islam, T., et al.** (2026). "FUME: Fused Unified Multi-Gas Emission Network for Livestock Rumen Acidosis Detection." *WACV 2026* (Oral)
- **Islam, T., et al.** (2025). "CarboFormer: Lightweight Semantic Segmentation for CO<sub>2</sub> Detection." *ISVC 2025* (Oral)
- **Islam, T., et al.** (2025). "WeedSwin: Hierarchical Vision Transformer with SAM-2 for Weed Detection." *Scientific Reports* 15(1), 23274
- **Islam, T., et al.** (2024). "Breast Cancer Classification with Explainable AI." *Scientific Reports* 14(1), 8487

## Awards & Leadership

**Honors:** 3rd Place, Illinois Young Innovator of the Year (Top 3 of 15 finalists, Falling Walls Lab Illinois 2025) | Associate Editor, *Frontiers in Medicine* (Q1 journal, youngest appointee) | ACM Professional Member | Reviewer, *Scientific Reports*

**Leadership:** General Secretary, Bangladesh Student Association (secured Best RSO Award 2025 for 150+ member organization) | Captain, Cricket & Badminton Championship Teams (2024-2025)